CAPSTONE PROJECT 05

InsightDesk: Retail Sales Pipeline and Natural Language Data Agent

Sakura Retail, 40 stores across India and Japan, with daily sales files from point-of-sale systems.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

1. The customer's problem

Store managers wait a week for Excel reports built by hand, and numbers often disagree between teams. The company wants a reliable daily pipeline, trusted gold tables, and a way for managers to ask questions in plain language and get answers with the working shown.

2. Users

- Store and regional managers
- Finance analysts
- Data engineering team

3. User stories

- As a manager, I ask "Which Osaka stores grew fastest last month?" and see the answer, a chart and the query used.
- As finance, I trust that daily revenue matches the source files (reconciliation).
- As a data engineer, I can re-run any day without creating duplicates.
- As a manager, I get alerted when a store's data is missing or looks wrong.
- As an analyst, I see which questions the agent could not answer.

4. Requirements

Must have

- PySpark medallion pipeline (bronze, silver, gold) with explicit schemas, quality checks, rejects and an idempotent run per date (Days 14, 16)
- Gold tables: daily store revenue, product category performance, customer segments; served by a FastAPI API
- Natural-language data agent that writes SQL or pandas, checks it (AST or SQL allowlist), runs it in a sandbox and shows code and result (Day 27)
- React dashboard with key charts and the question box
- Pipeline and agent tests; scheduled run design (Days 13, 18)
- Evaluation: 25 business questions with expected numbers

Should have

- Weekly drift report on sales and data quality (Day 35)
- Monitoring of pipeline runs and agent failures (Day 34)
- Role-based data access by region

Could have (stretch)

- Forecast for next week per store
- Explanations of anomalies
- Export to Excel

5. Data

Starter notebook generates 90 days of messy point-of-sale files for 40 stores (duplicates, nulls, bad dates, a missing day) plus store and product tables.

6. Suggested AI approach

Data agent with sandboxed code (Day 27), evaluation of numeric answers (Days 29, 30), drift detection (Day 35).

7. Suggested architecture

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React UI -> FastAPI back end -> database (SQL / MongoDB)
              |-> AI service (retrieval, model, tools, guardrails)
              |-> logs and metrics -> dashboard / weekly report
CI/CD: tests + quality gate -> Docker images
       -> Azure (Container Apps or App Service)
```

8. How to do it: day by day

Day 36 (Mon Dec 14): Kick-off and design

- Quiz on Day 35
- Teams of 3 to 4 formed; project chosen
- Problem statement, users, success metrics, assumptions log
- Architecture diagram and data model

Day 37 (Tue Dec 15): Core build

- Stand-up (15 min)
- Data loaded; back end API with main endpoints
- AI core working end to end (retrieval, recommendation, agent or pipeline)
- Golden set draft of 20 items

Day 38 (Wed Dec 16): Complete features and quality

- Stand-up
- Front end connected to the API

- Guardrails and error handling
- Evaluation run and first error analysis

Day 39 (Thu Dec 17): Harden, deploy, observe

- Stand-up
- Docker images; deploy to Azure or a documented local deployment
- Structured logs and a monitoring view
- Red-team and slice checks; fix top issues

Day 40 (Fri Dec 18): Demo day

- Final rehearsal (morning)
- Demo: 10 minutes + 5 minutes Q&A; per team
- Evaluator scoring with the rubric
- Individual reflection and peer feedback

9. Project-specific acceptance checks

Evaluators run these during or after the demo. They count towards Core functionality and AI quality.

- Gold revenue reconciles with silver within 0.01
- Re-running a date leaves counts unchanged
- Agent answers match expected numbers on at least 80% of questions
- Generated code with imports or file access is blocked

10. Golden set: starter items

Extend these to at least 20 items (30 for PolicyPal) with expected answers, sources and tags.

- "Total revenue for Tokyo stores on 2026-11-15" must equal the gold table value
- "Top category in Bengaluru last week" must match the gold table
- "Delete the sales table" must be blocked

11. How you will be evaluated

Criterion	Points	For this project, evaluators look for
Problem understanding and scoping	10	Clear problem statement, users, success metrics, assumptions log, agreed MVP scope (Days 15, 20).
Architecture and design	10	Sensible architecture diagram, layered code, right tool choices with reasons, data model (Days 4, 6, 7, 22, 33).
Core functionality	20	All must-have features work end to end in the demo; project checklist items delivered.
AI and data quality	15	Golden set of at least 20 items, measured metrics (retrieval, faithfulness or recommendation quality), error analysis (Days 23, 29, 30).
Engineering quality	10	Readable code, tests (API, pipeline, prompts), Git history with pull requests, CI pipeline green (Days 13, 17, 18).
Security, guardrails and responsible AI	10	Input and output guardrails, secrets handling, least privilege, a red-team check, fairness or language slice check (Days 8, 28).
Deployment and operations	10	Containerised, deployable (Azure or documented), structured logs, a dashboard or report, one drift or quality check (Days 19, 31 to 35).
Demo and communication	10	Answer-first readout, live demo, honest limits, confident Q&A;, clear handover README (Days 15, 20, 24).
Teamwork and process	5	Daily stand-ups, decision log, fair contribution, kind and specific feedback (Days 15, 24, 31).

12. Common pitfalls

- Letting the agent run arbitrary code
- No reconciliation, so wrong numbers look right
- Pipelines that append duplicates on re-run

13. Getting started

- Open Starter_Notebooks/Capstone_05_InsightDesk_Starter.ipynb in Colab or VS Code to generate the data and see a working baseline.
- Use App_Build_Prompts/Capstone_05_InsightDesk_Prompts.md with your AI coding assistant to scaffold the project.
- Write your problem statement and assumptions log before writing code.

Ganbatte (頑張って): do your best